OMNI CAMERA ROVER | REV C PROTOTYPE

1. Controller

File: controller.kicad_sch

2. Motors

File: motors.kicad_sch

3. Charging

File: charging.kicad_sch

4. Regulator

File: regulator.kicad_sch

5. Interlock

File: interlock.kicad_sch

Battery protection

File: battery_protection.kicad_sch

PWR_FLAG GND

PWR_FLAG VMOTOR

PWR_FLAG USB_5V

PWR_FLAG U2_VINT

PWR_FLAG U3_VINT

Editable functional sheets. Global net names link sheets.
Prototype: electrical, thermal and mechanical qualification remains open.

Flex camera

File: camera.kicad_sch

Nathaniel Sanford

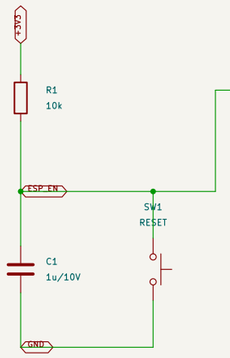
Sheet: /
File: omni_ugv.kicad_sch

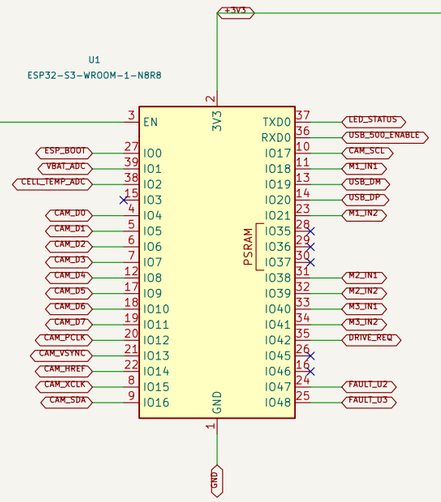
Title: Omni UGV / Rev C

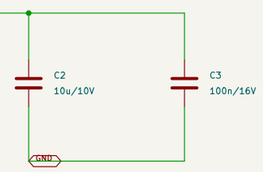
Size: A3	Date: 2026-11-06	**Rev: C-PROTOTYPE**
KiCad E.D.A. 10.0.8		Id: 1/8

Controller, camera and service points

RESET / enable RC







Waveshare OV2640 | documented 2 x 9 interface

Status indicator



BOOT strap



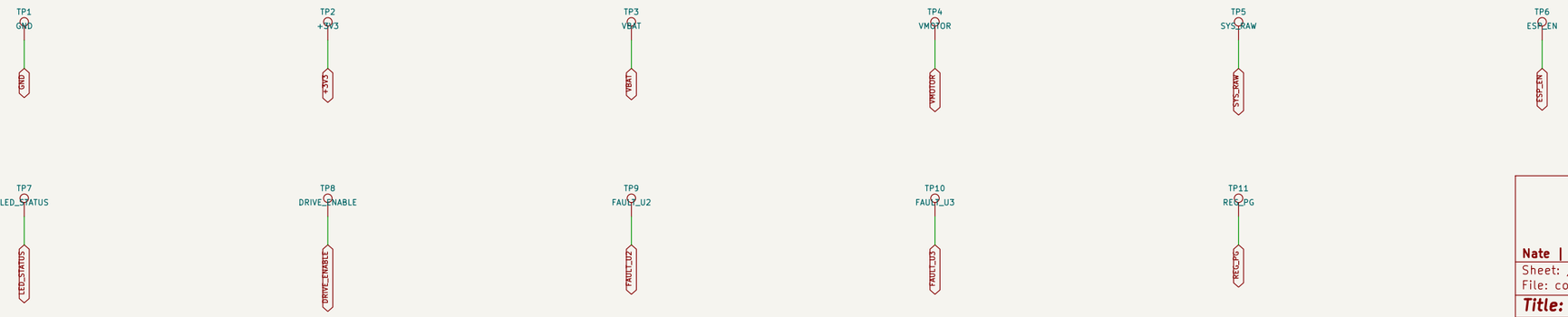
Camera reset RC





[SPEC] GPIO35 / 36 / 37 reserved for N8R8 PSRAM.

Service points (named nets continue on other sheets)



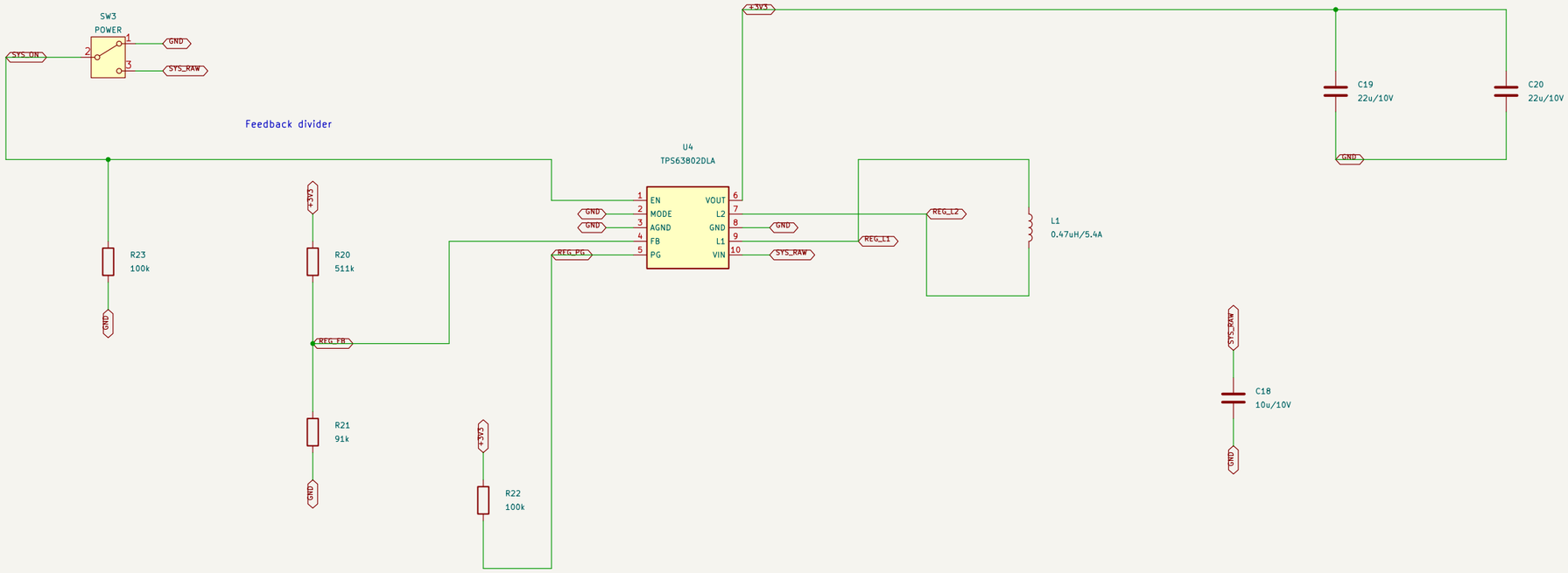
Nate | Omni camera rover		
Sheet: /1. Controller/		
File: controller.kicad_sch		
Title: Controller, camera and service points		
Size: A3	Date:	**Rev: B – prototype**
KiCad E.D.A. 10.0.8		Id: 2/8

3.3 V buck–boost supply

System enable

Feedback divider

Output capacitors

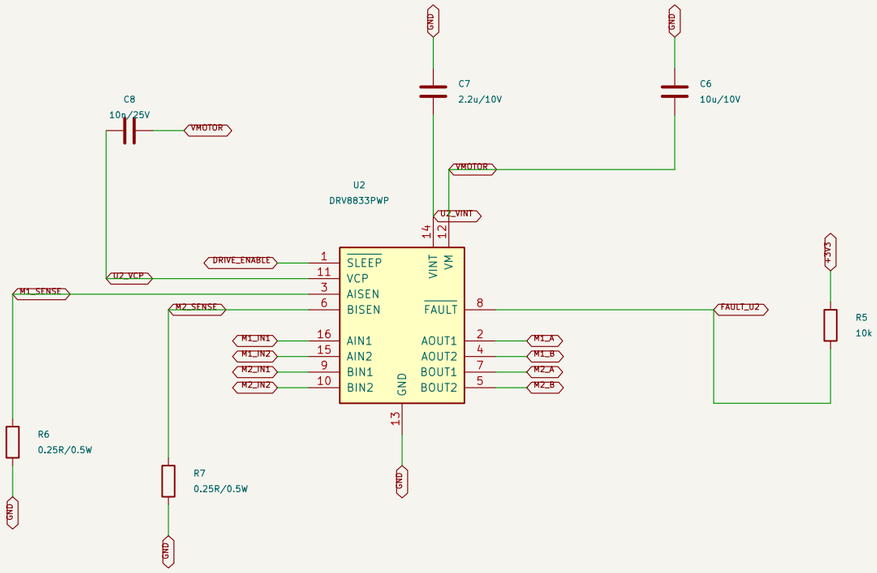


[TARGET] Keep switching loops short. Place input/output capacitors at U4.
[TBD–MEASURE] Verify startup, load transients and capacitor effective capacitance.

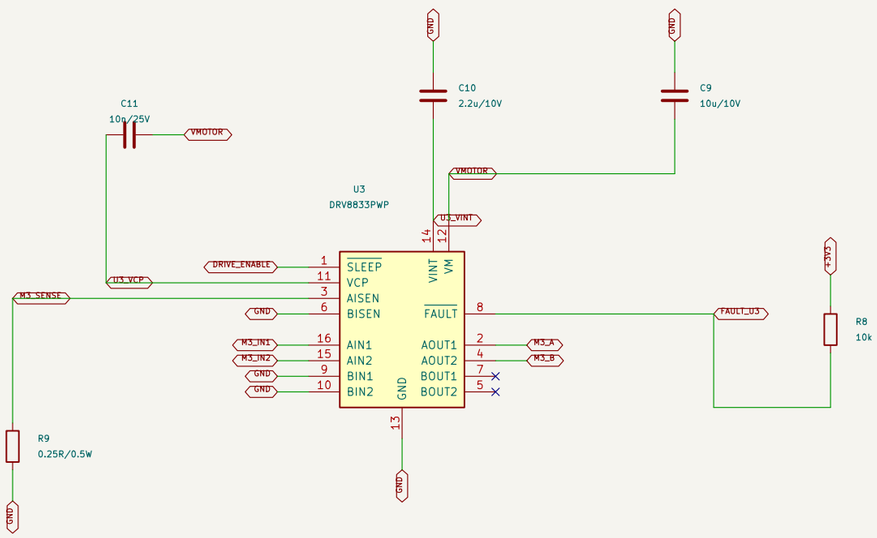
Nate | Omni camera rover		
Sheet: /4. Regulator/		
File: regulator.kicad_sch		
Title: 3.3 V buck–boost supply		
Size: A3	Date:	**Rev: B – prototype**
KiCad E.D.A. 10.0.8		Id: 3/8

Three independent motor channels

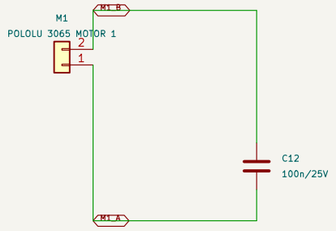
U2: wheels 1 and 2

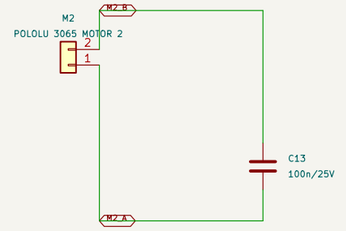


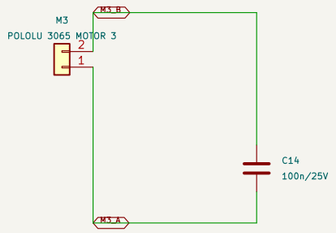
U3: wheel 3; unused bridge pins retained as designed

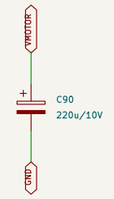


Motor terminal pads and suppression capacitors







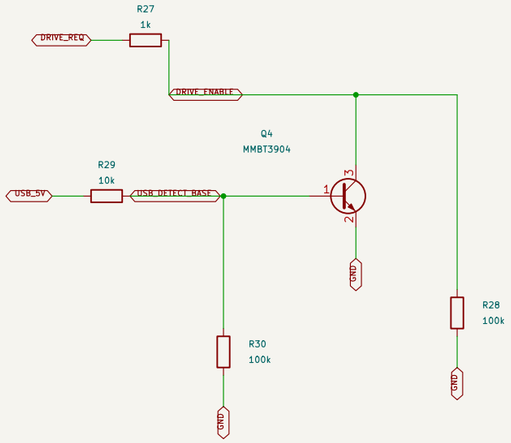


[TARGET] Route motor current away from logic returns.
[TBD-MEASURE] Braking overshoot, driver temperature and carpet load.

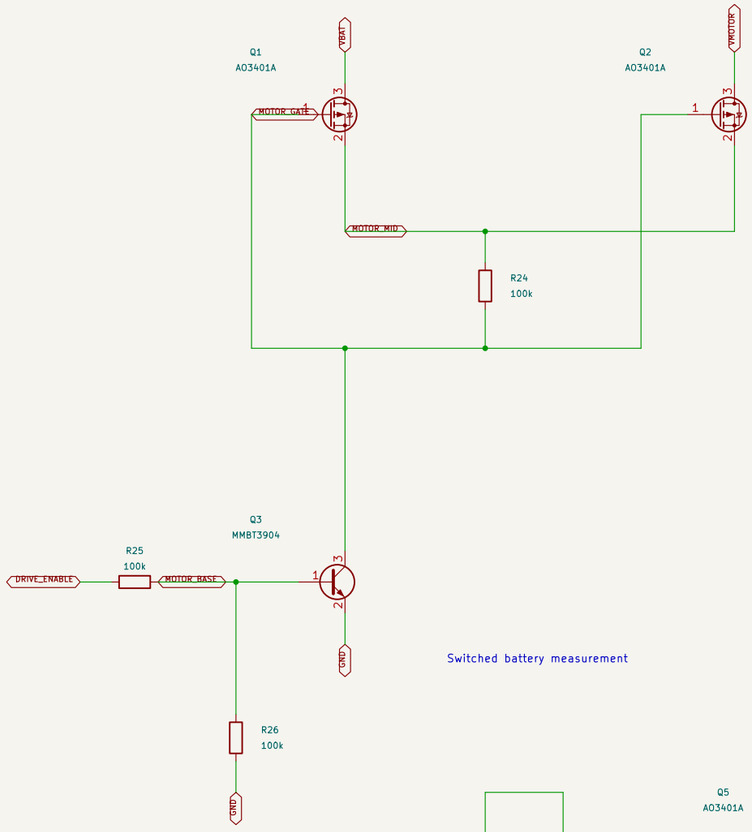
Nate | Omni camera rover		
Sheet: /2. Motors/		
File: motors.kicad_sch		
Title: Three independent motor channels		
Size: A3	Date:	**Rev: B — prototype**
KiCad E.D.A. 10.0.8		Id: 4/8

USB motor interlock and battery measurement

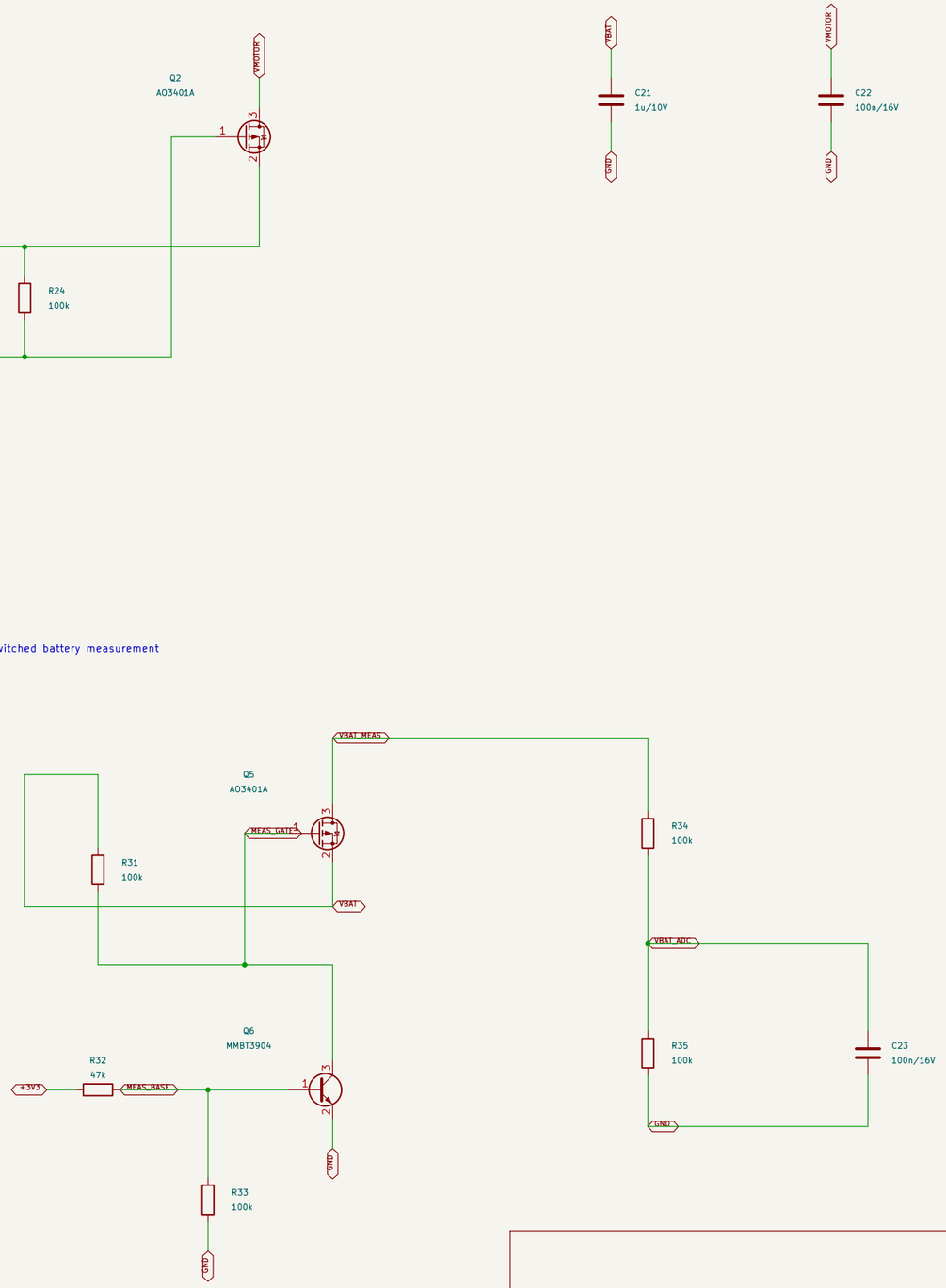
USB presence clamps DRIVE_ENABLE



Back-to-back motor supply switch



Switched battery measurement



[TARGET] Reset / absent commands keep motors disabled.
[TBD-MEASURE] Verify USB insertion/removal and motor braking transients.

Nate | Omni camera rover

Sheet: /5. Interlock/
File: interlock.kicad_sch

Title: USB motor interlock and battery measurement

Size: A3	Date:	**Rev: B – prototype**
KiCad E.D.A. 10.0.8		Id: 5/8

USB input, charging and cell connections

USB data to ESP32 native USB

J1
USB-C
VBUS A4
CC1 A5
CC2 B5
D− A7
D− B7
D+ A6
D+ B6
SBU1 A8
SBU2 B8
SHIELD S1
GND A1
USB_5V
USB_CC1
USB_CC2
USB_DM_EXT
USB_DP_EXT
GND

R10
5.1k
R11
5.1k

U6
USBLC6-2SC6
USB_5V
USB_DM_EXT
USB_DP_EXT
GND

R12
22R
USB_DM
R13
22R
USB_DP

Raw cell –> protection sheet

J2
RAW 1S CELL 4.2V MAX
CELL_N
CELL_P

Charge / system power path

USB_5V
SYS_RAW

C15
1u/10V
C17
10u/10V
GND

U5
BQ24074RGT
ITERM
CE
TMR
EN1
EN2
ILIM
ISET
IN
OUT
BAT
TS
PGOOD
CHG
VSS
USB_500_ENABLE
CHG_TIMER
CHG_ILIM
CHG_ISET
VBAT
BAT_NTC
CHG_PGOOD
CHG_STATUS

C16
10u/10V

R16
46.4k
R17
100k
R14
3.57k
R15
3.24k
R19
100k
+3V3
CHG_PGOOD

R18
2.2k
D2
CHARGE AMBER
SYS_RAW
CHG_LED_A
CHG_STATUS

Cell-bonded charger probe

J4
CELL NTC – CHARGER
GND
BAT_NTC

[SPEC] J2 CELL_N is not system GND: R36 is the only intended return path.
J4: independent insulated two-wire 103AT-2 probe on the cell.
Motor current uses protected VBAT, not charger SYS_RAW output.

Nate | Omni camera rover	
Sheet: /3. Charging/	
File: charging.kicad_sch	
Title: USB input, charging and cell connections	
Size: A3	Date:
KiCad E.D.A. 10.0.8	**Rev: B – prototype**
	Id: 6/8

1S protection and cell-temperature monitoring

Cell voltage / current sensing

R37
100R 1%

C24
100n/16V X7R

R38
20R 1%

U7
BQ298217RUGR

R36
10m 1% 1W

R39
1k 1%

Common-drain high-side protection FET pair

R41
10M

Q8
CSD17308Q3

Q7
CSD17308Q3

R40
10M

TH2 -> J5 -> GPIO1 / ADC1

R42
10k 1%

J5
CELL NTC - DRIVE

R43
1k 1%

TH2
103AT-2 CELL-MOUNTED

C25
100n/16V X7R

TH1 -> J4 charger input

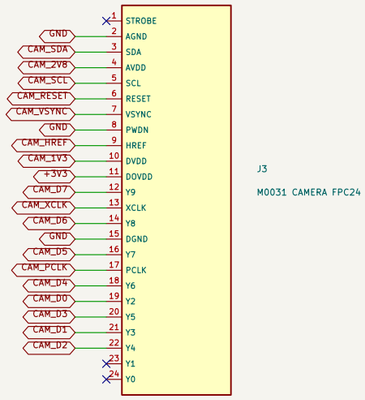
TH1
103AT-2 CELL-MOUNTED

#FLG20
PROTECTION_PWR_FLAG

#FLG21
PROTECTION_PWR_FLAG

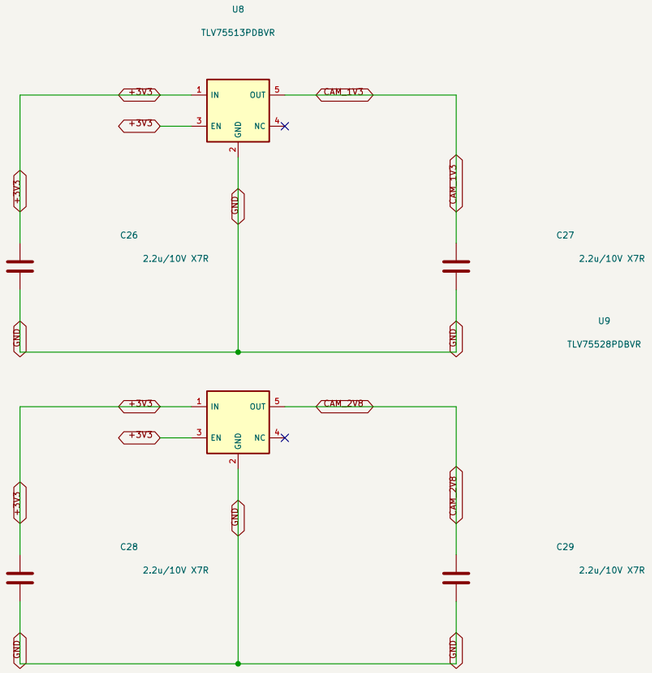
[SPEC] TH1 and TH2 are OFF-BOARD, bonded to the cell through electrical insulation.
Separate two-wire probes; do not connect either sensor to raw CELL_N.
[TARGET] Firmware must stop driving on overtemperature, open/short probe or stale readings.

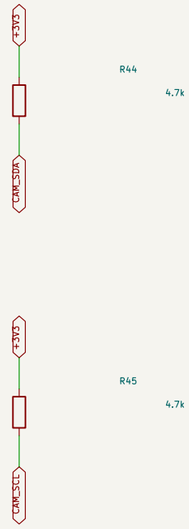
Nate | Omni camera rover	
Sheet: /Battery protection/	
File: battery_protection.kicad_sch	
Title: 1S protection and cell-temperature monitoring	
Size: A3	Date:
KiCad E.D.A. 10.0.8	**Rev: B - prototype**
	Id: 7/8



J3

M0031 CAMERA FPC24





M0031 only. Verify flex contact side, thickness and pin-1 before assembly.
1.3 V core / 2.8 V analog / 3.3 V I-O. No interchangeable generic OV2640 pinout.

Sheet: /Flex camera/	
File: camera.kicad_sch	
Title: OV2640 M0031 flex interface	
Size: A3	Date:
KiCad E.D.A. 10.0.8	Id: 8/8